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Quantitative analysis

by Mark Franklin

(Early draft)

When we quantify we attempt to employ the scientific method to discover things about the world. In the social sciences we are trying to discover things about the social world, but the approach is still scientific. The scientific approach attempts to abstract from the nuances and details of a story the salient features that can be joined together into a theoretical statement (or statements) expected to hold true of any situation in which the same salient features can be abstracted. If such a statement does not hold true in some specific situation this is presumed to be either because the theory was wrong or because it was not sufficiently elaborated. Elaborating social theories to bring in additional features of the world, found necessary for a full explanation, is an important feature of the scientific approach .

The transition from case studies to quantitative analysis is largely a matter of the number of cases. If you have one case, no causal inferences can be made. If you have two cases you can rule out something as a cause of something else. If you have three cases you can rule out two things, or you can start to make quantitative statements (for example, something might be found to pertain two thirds of the time). As soon as you start saying things like “this happens two thirds of the time” you are doing quantitative analysis. But in order to make such statements you need to be able to abstract general features that are common to many cases, as already mentioned. You also need a fairly large number of cases in your study.

Exactly what constitutes “fairly large” in the above statement is not at all clear and in practice there is a large area of overlap in which one researcher would talk of a “multiple case study” while another would talk of a “small N study” (the letter N in the quantitative tradition stands for “number of cases” and as soon as you see cases referred to in that way you know you are reading something written in the quantitative tradition).

So whether you do case studies or quantitative studies depends, over a large area of overlap, on what tradition you are working in rather than on what you are doing. Consider the following example:

Table 1

	Multiple social networks going back to C12	Lack of social networks even today
High quality democratic Governance	Northern Italy	
Poor governance		Southern Italy

(Adapted from Robert Putnam, *Making democracy work* – Princeton University Press 1993)

This rather famous example was put into question only a few years later by a Harvard PhD thesis which looked at French regions and found cases of poor governance even where there were longstanding social networks. What do do? One possibility would be to conduct additional studies in the hopes of discovering that either the French or the Italian cases were happenstansical – so unusual as to be not worth worrying about. One might, after a lot of work, come up with a table that looked something like the picture shown in the table below, where 2 cases of high quality governance and 3 cases of poor governance match the Italian case while the exception found in that thesis I mentioned turns out to be the only one:

Table 2

	Multiple social networks going back to C12	Lack of social networks even today
High quality democratic governance	2	0
Poor quality governance	1	3

That seems pretty definitive: the Italian case holds far more often than not. Moreover we appear to have discovered that multiple social networks are a necessary, even if not a sufficient, condition for high quality governance.

It would, of course, be far more interesting to discover WHY the exception occurred, which would mean using the additional cases to see whether some other condition accounted for the exception. If we could find a magic ingredient (call it entrepreneurship) that accounted for the difference we could make a table that looked like

Table 3

	Entrepreneurship		No entrepreneurship	
	Multiple networks		Multiple networks	
	Yes	No	Yes	No
High quality governance	2	0	0	0
Poor quality governance	0	1	1	2

Now we see that there are actually two conditions, both of which must be present, for high quality democratic governance. The condition found necessary by Putnam, and also an additional condition that he knew nothing about and which turns out to be responsible for the exceptional French case. The additional condition turns out to be a second necessary condition for high quality governance (entrepreneurship without multiple networks does not yield high quality governance any more than does multiple networks without entrepreneurship).

Let me put the actual names of the regions concerned into a simpler table where the two conditioning variables determine where each region appears in the table, and the quality of governance in each region is indicated by a tick or a cross. This table is the sort of table one would expect to see in a multiple case study, whereas the table above is the sort of table one would expect to see in a quantitative analysis (here I use the terminology of the author of the thesis)…

Table 4

	Territorial policy communities (multiple social networks)	
	Yes	No
Political entrepreneurship	Brittany √ Tuscany √	Languedoc ✗
Lack of entrepreneurship	Aquitaine ✗	Provence ✗ Liguria ✗

(Marc Smyrl, *European policies, regional programs, local politics* – Harvard PhD dissertation 1997)

Of course, with only six cases it is hard to be sure that one has exhausted the possibilities. Additional exceptions may lurk around the next corner, and additional conditions might need

to be taken into account. But it is pretty obvious that to discover more one would need a great many additional cases, and with a great many additional cases the format used in Table 3 becomes more useful than that in Table 4. If we had hundreds of names in each list the information would not be very useful if presented in the second format. Quantification requires abstraction, as already mentioned.

Beyond that area of overlap, still within the range of "small N studies" a researcher requires quantitative methods simply in order to be able to talk about his or her findings. With more than about ten cases it becomes necessary to use numbers to summarize what you have learned. But with small N studies what you can say with numbers is still quite limited. From this perspective, the next important watershed comes with the transition to 'large N studies' where you can bring to bear the full power of what is called 'multivariate analysis'. Again there is no fixed boundary. Small N studies shade into large N studies at somewhere between thirty and three hundred cases.

So what can be done with small N studies that cannot be done with case studies, and what can be done with large N studies but not with small N studies? Essentially we can say that, as the number of cases goes up, so the researcher has a better shot at being able to

a) specify the conditions under which causal effects are felt (how widespread they are)

b) specify the nature of the causal effects (how strong they are)

c) specify how likely it is that the effects are real rather than happenstansical (how significant they are).

<u>The vocabulary of quantitative research</u>

The distinction I just made (between widespread, strong, and significant causal effects) brings us face to face with the main difficulty involved in quantitative analysis. In order to be able to talk quantitatively one has to be able to make distinctions that to most people do not come naturally. We have to be able to use a distinctive vocabulary that to most people sounds like jargon. Many of these distinctions, and the words used to make them, sound rather arbitrary. In ordinary English the difference between strong, widespread, and significant is not obvious. All three words sound pretty synonymous to the untutored ear. All appear to be variants on the word "important".

Well, that is true. They are all variants on the word "important". But, as with the fifty different words that Eskimos have for "snow" distinctions that appear unimportant from some points of view can seem very important from other points of view.

In brief, a presumed causal effect is a strong one if it appears to cause extensive change. It is widespread if it occurs in many different circumstances and situations, and it is significant if it is unlikely to be spurious or happenstansical. An effect that is unlikely to be the just a chance phenomenon may yet be quite small or quite restricted in its applicability. Or it might be widespread but small, and so on. These words are widely used in talking quantitatively about the world. When talking about accidents we use much the same vocabulary, distinguishing between a freak accident that probably will never happen again and one that is significant because it is part of a predictable pattern. But even a significant accident might have small or restricted consequences. Alternatively its consequences could be major and/or widespread.

There is quite a lot of vocabulary to be learned in order to be able to talk sensibly about quantitative social research findings (or in order to be able to make sense of the literature that makes use of this vocabulary). In the rest of this chapter I will go through some of the more important words in this vocabulary. Evidently learning to do quantitative social research involves somewhat more than just learning the vocabulary. There are some corresponding skills, but I have always found that it was the vocabulary that got people hung up rather than the skills. You may find it helpful to take a sheet of paper and write down the words in quotation marks that follow, so as to have a crib sheet to use as you move forward.

Sources of quantitative information

Quantitative information can be collected in exactly the same way as any other information – by means of interviews (generally called ‘surveys') or by looking it up in compendia of various kinds (or on the web). Although there is no logical reason why this should always be true, surveys generally involve sampling (because there are generally too many individuals for us to survey them all) whereas information that we look up is generally exhaustive (we can obtain data for all the cases that interest us). It is important to know whether the information was gathered by means of a sample, because samples are subject to error when we try to generalize beyond the sample. This of course is equally true for many case studies, where the possibility of an 'unrepresentative case' is synonymous with a 'bad sample', but there are certain types of samples ('random samples') for which it is possible to use statistical methods to generalize beyond the sample with a known probability that the generalization will be true. This is a very powerful feature of random samples that is unavailable to those who select their cases in other ways. Most surveys are based on random sampling. Though

there are different types of random samples, which need to be distinguished in practice, such distinctions are beyond the scope of this introduction.

The dataset and data matrix

As soon as one starts talking about quantitative information one is forced to start talking about data. Data (the word is plural – treating data as a collective noun is common but wrong) arise from standardized information. In this sense, a biographical compendium contains data, because the characteristics of each individual are presented in a standard form: gender, birthdate, schools attended, etc. A dataset goes further in coding the standardized data, generally in numerical terms (eg 1=female, 2=male) and providing a dictionary or 'codebook' with which to interpret the codes. When organized in this way, the codebook is conceptually distinct from the 'data matrix' which is a rectangular table organized with different cases in different rows. Across the table are columns, each column containing a particular characteristic (gender, age, income, party voted for, etc.). These are known as 'variables'. By looking at the intersection of a particular row with a particular column one can

Table 5

Data matrix

country	electn	EPturnt	natturnt	yrsleft	compuls	first
bri	1979	32.2	76	4	0	1
den	1979	47.8	86	0.36	0	1
bel	1979	90.4	95	2.4	1	1
fra	1979	60.7	83	2	0	1
ger	1979	65.7	91	1.3	0	1
ire	1979	63.6	76	2	0	1
ita	1979	84.9	91	4	1	1
lux	1979	88.9	89	0	1	1
net	1979	57.8	88	2	0	1
bri	1984	32.6	73	3	0	0
den	1984	52.4	88	3.2	0	0
bel	1984	92.2	95	1.3	1	0
fra	1984	56.7	71	1.7	0	0
ger	1984	56.8	89	2.6	0	0
gre	1984	82.2	82	0.96	1	0
ire	1984	47.6	73	2.7	0	0
ita	1984	83.4	89	3	1	0
lux	1984	88.8	89	0	1	0
net	1984	50.6	81	1.9	0	0
por	1984	72.4	73	0	0	1
spa	1984	68.9	70	2.4	0	1

Codebook

Variables	Meaning (and values)
country	Three-character country id
electn	Date of election (year)
EPturnt	Turnout at European Parliament election (percent)
natturn	Turnout at previous national election (percent)
yrsleft	Years to next national election (years and parts of yrs)
compuls	Compulsory voting at time of EP election (0=no; 1=yes)
first	First EP election held in country (0=no; 1=yes)

read off the particular characteristic or 'value' associated with a particular case. Thus if turnout at the European Parliament election were to be the variable in the third column of the table (as shown below) and France was the case in the fourth row, then by looking across the fourth row to the third column one would find that French turnout was 60.7% at that election.

Variables and levels of measurement

Talking about variables is complicated by the fact that there are different types of variables. Implicitly we have already mentioned two types: variables like gender, where the values ascribed are quite arbitrary, and variables like age, where the values ascribed have an intrinsic meaning (age is generally measured in years). In the case of a 'nominal' variable like gender, men could as easily be coded 1 and women 2 as the other way around – or the two possible values could be coded 'M' and 'F'. All we are doing with a nominal variable is distinguishing the characteristics that can apply to different cases in terms of that variable – the values we employ do no more than name the characteristics (hence 'nominal' from the Latin for 'name'). But with 'interval' variables like age the intervals between the values are meaningful (a year or a dollar, or some other 'unit of measurement'). Two more levels of measurement are important to social science researchers. Variables can be 'ordinal' if the values have an order that is implied by their numeric values (5 is bigger than 4) even if there is no unit of measurement, and they can be 'dummy variables' if all they do is indicate the presence or absence of some characteristic: 0=not British, 1= British (for example). When the data come from a survey of individual people, the most common variables are nominal and ordinal, whereas the variables we really want for analysis purposes are interval. A lot of time and skill is expended by researchers in 'transforming' their data so as to overcome this problem. The solution generally adopted in political science research is to treat ordinal variables as interval (provided they have enough categories) and to recode nominal variables into their dummy counterparts (which can be regarded as honorary interval variables with a unit of measurement that is the presence or absence of the attribute in question).

Table 6

Level of measurement	Example	Additional information contained
Interval variable	43% Lab; 10% Lib; 47% Con	Quantity (Con 4% more than Lab)
Ordinal variable	1=Labour, 2=Liberal, 3=Con	Order (left-right relative location)
Nominal variable	1=Labour, 2=Con, 3=Liberal	Mutual exclusivity
Dummy variable	0=Not Labour; 1=Labour	n.a.

In the above table, not only do we see examples of different types of variables but we also see a summary of the additional information needed to code a variable at a higher level than the level below it in the table (this is also the additional information imparted by such a coding). Dummy variables can be thought of as having the lowest level of information – the presence or absence of an attribute. Descriptions made in ordinary language generally consist of strings of attributes ("the man has blue eyes"). Talking in terms of attributes enables us to string together different attributes of the same type ("the man has one blue eye and one green eye"). As soon as we move up to the nominal level we assert that the attributes are mutually exclusive (one is allowed to vote only for a single political party, so a code of Conservative implies not Labour and not Liberal). By taking an additional step to the ordinal level we introduce some additional concept that enables us to order the values – and also introduces the possibility of miscoding the variable according to this concept, as in the example in the table, where commentators argue about whether Labour and Liberal have recently swapped places in left-right terms.

Talking about variables requires us to make one further distinction, between variables whose behaviour we are trying to understand (so-called 'dependent' variables) and variables we are using in order to arrive at the required explanation (so-called 'independent' variables). In the example we used earlier, quality of democratic governance was the dependent variable because we were trying to answer the question "what does the quality of governance depend on? " Extensiveness of networks and the availability of entrepreneurial talent were independent variables because we were not (in that analysis) asking what they depended on. (Note that in some other piece of research one or other of those variables might very well be treated as dependent if, for example, we wanted to know what the availability of entrepreneurial talent depends on).

Units and levels of analysis

Qualitative as well as quantitative analysis happens at many different levels. One may analyse countries, years, regions, cities, schools, people, or events – and much more. These are referred to as the units of analysis, or cases. The number of cases is referred to by the capital letter N, as already mentioned. Units of analysis can be distinguished in terms of the level of analysis at which they fall: the national unit is at a higher level than the city unit which in turn is at a higher level than the individual who lives in that city and country. In the above table, the example given of an interval variable is of a variable measured at a higher

level not only of measurement but also of analysis. To be able to say that Labour received 43 percent of the vote one has to be talking about an aggregation of individuals (most likely all of those voting at a particular election in a particular country) rather than of a particular individual. Because higher levels of analysis so often involve information about multiple individuals, the data concerned are often referred to as 'aggregate data'. The other examples in the table are ambiguous as to level of analysis (they could refer to political parties as easily as to individuals) but it is likely that they are variables measured at the individual level of analysis.

Though it is possible to investigate research questions that involve units at different levels of analysis, it is important to be clear about how these units are related to each other. This is just as true in qualitative as in quantitative analysis, but it generally goes without saying that one cannot compare a city with a school (for example). With large N studies it is easier to become confused about the level of analysis of different components of the study. But the most important thing to realize about the level of analysis is that the type of variables we find at different levels tend to be different. I already mentioned that with individual-level data we get very few interval variables (in order to find an example of an interval level variable relating to parties I had to move up to an aggregate level of analysis). In addition, individual-level data contains a huge amount of error or 'noise'. People make mistakes when answering survey questions or when filling in forms. People fail to understand the questions they are asked or the meaning of the answers that they give. Most importantly there is always a disjunction between the person who designs the questions (and hence the coding scheme for those questions) and the person who answers them (thus implicitly providing the values that will be coded), For this reason the questions often fail to communicate exactly the meaning intended. All of this results in error. There is much less error in higher-level data (because the people who design the coding scheme for such data are generally the same as the people who code it). We are also much more likely to find interval variables in aggregate data because the very act of aggregation yields variables that count the number (or proportion or percentage) of individuals in different categories or with different characteristics. The percentage voting conservative (an aggregate phenomenon) is very definitely an interval variable, whereas the same variable at the individual level (voted conservative) is a nominal variable, as we have already seen.

Statistics

Now we get to the topic that really puts spots before people's eyes. In order to talk about the findings of quantitative research one needs to use statistics. Technically speaking, statistics are 'coefficients' that summarize things of interest about data. Statistics are also the procedures by which one arrives at such coefficients (generally referred to by the cogniscenti as 'statistical analysis'). A percentage or an average is a statistical coefficient (generally referred to as a 'descriptive statistic' because it describes a body of data), but much more interesting to social scientists are coefficients that address the questions summarized earlier: How widespread? How strong? How significant? We will start with the last of these.

How significant?

'Significance' relates to the chances of being wrong when making some assertion. Statistical methods allow us to determine the chances of being wrong when we are dealing with a random sample. By extension most researchers (led by economists, who have the strongest credentials for methodological purity) apply these methods to any dataset for which there is no reason to doubt its representative nature. Questions of significance can be applied to what are called 'point estimates' (what are the chances that we are wrong when we predict that the Democrats will win more than 50% of the two-party vote in the 2008 US Presidential election?), but much more interesting to social scientists are questions about the significance of a relationship between variables. If we take the example used earlier, of the relationship between the extent of policy networks and the quality of governance, it would be worth knowing what the chances are that the relationships found by Putnam and Smyrl are significant ones – that is, that they are unlikely to be the result of happenstance and are thus likely to be found again and again as we look at other regions and countries.

Whether a relationship is significant depends on three things:

1) the strength of the relationship;
2) the number of cases investigated when establishing the relationship; and
3) the degree of certitude required before we are willing to accept a statement as true

Starting with the last of these, if we require 100% certitude it will follow that no relationship is significant. The industry standard in the social sciences is to accept a statement as true if it is likely to apply 95 percent of the time (which is the same as saying that it will be wrong five percent of the time – for which reason it is referred to as 'significance at the 0.05 level'). Note that this is not a very stringent test. If five percent of cases will fail to show the relationship

concerned, this means that one in twenty cases will fail to show the relationship. If we count up all the journal articles reporting findings significant at the 0.05 level, the number that are probably wrong amounts to hundreds or even thousands! Unfortunately we do not know which ones those are, except to the extent that some of the findings have been replicated in other research using other data (and preferably other methods as well). In the social sciences, as in the natural sciences, the real test for the truth of an assertion comes from the fact that it can be replicated. Enough different findings of the same thing reduces the probability of being wrong to infinitesimal proportions. The great thing about scientific statements is that they have resisted attempts to falsify them. This is as true of social sciences as of natural sciences. (Of course, the same logic regarding significance implies that one in twenty attempts at falsifying an assertion can succeed at the 0.05 level of significance without this meaning that the assertion concerned is actually false).

The second thing that determines whether a relationship is significant is the number of cases employed when arriving at the assertion regarding that relationship. If we want to be able to assert that there is a significant relationship between the extensiveness of social or policy networks and the quality of democratic governance, the more cases we have investigated in arriving at that assertion the better. Enough cases can render any relationship significant at any non-zero level of significance, so with enough cases the question of significance ceases to be very interesting. In the early years of social research, investigations of district-level data in India was popular because, no matter what the relationship, it was always significant simply because India contains so many districts. Nowadays we realize that, to be interesting, it is not sufficient for a relationship to be significant, it must also be a strong (or powerful) one. But it is also important to realize that, even with a relatively small N, relationships can prove significant if they are strong enough.

How strong?

To determine how strong a relationship is, we must determine the amount of change in the dependent variable that is brought about by change(s) in the independent variable(s). If the chances of a conservative victory go up by 4 percent when its leader is a popular figure (rather than an unpopular one) that is not as strong a relationship as if the chances of a conservative victory go up by 40 percent in such circumstances. A small change is much more likely to be happenstansical than a large change, but more importantly a small change is not very interesting even if it were to prove significant (as already mentioned). When talking

about strengths of relationships it helps to think in terms of a graph that has the dependent variable arrayed up and down the vertical axis and an independent variable arrayed along the horizontal axis. For a given movement across the horizontal axis we can then read off the corresponding movement up the vertical axis, as follows:

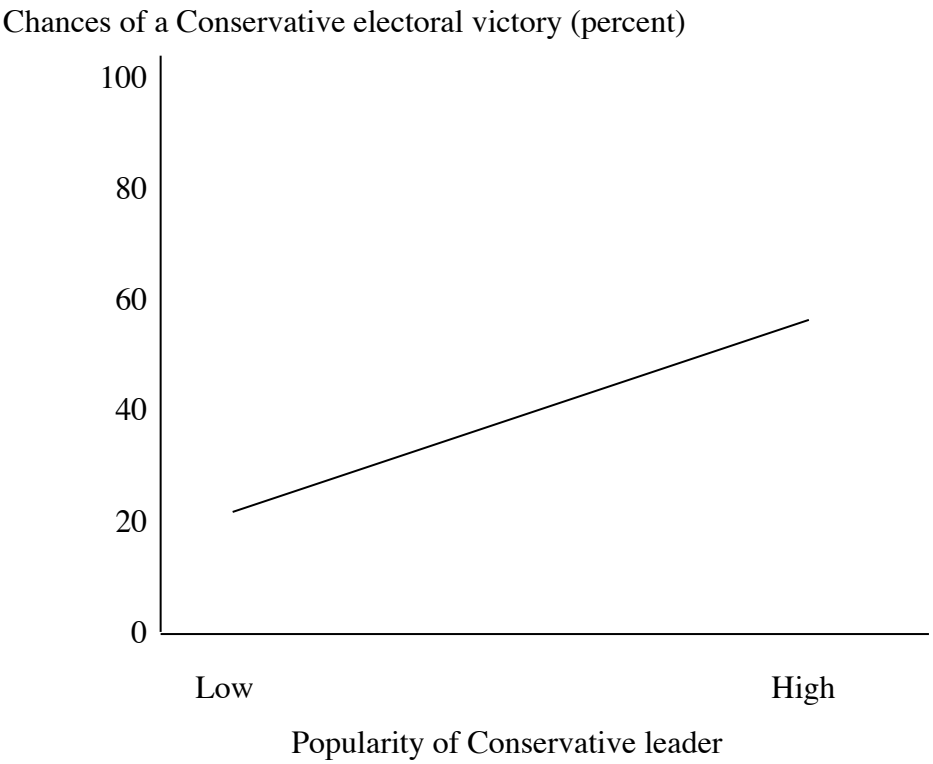

Figure 1

In this graph we see the chances of a Conservative victory increasing from only 20 percent to about 60 percent as the popularity of the Conservative leader increases from low to high. This corresponds to a 40% difference (60 – 20, or an 'effect' of 0.4, since effects are generally expressed as proportions). One can think of the slope of the line in terms of the leverage it shows the independent variable having on the dependent variable. An almost flat line corresponds to very little leverage. A strongly sloping line corresponds to much more leverage. An effect of 0.4 gives quite a lot of leverage. By contrast, it is clear that an effect of only 0,04 (4%) would yield a line that was almost flat – a line with almost no leverage. (A downward slope is also possible, and would indicate a negative relationship: higher values of the independent variable would correspond to lower values of the dependent variable).

The relationship shown in a table (such as those we used earlier) can easily be converted to a graph such as the one above by percentaging the table in the direction of the

dependent variable. Thus in Table 2 above (the first of those relating to Putnam's theory that contained any numbers), the dependent variable (quality of governance) runs down so we percentage down, and find that 67 percent of regions with extensive social networks (2 out of 3) see high quality governance whereas 0 percent of regions without extensive networks see high quality governance. Subtracting we find that social networks make a difference of 67 – 0 = 67 percent to the quality of governance (social networks have an effect on governance of 0.67). That is a pretty strong effect on a scale that goes from 0 to 1 which, if turned into a graph, would show a slope even steeper than the one depicted above. The steepness of the slope in this example corresponds to our intuition that a single exception to Putnam's asserted rule does not amount to much, but the small number of cases certainly would preclude even so strong an effect from being statistically significant.

Correlations between variables

At this point we need to take a brief detour to talk about correlation analysis. Though the political science journals hardly mention correlations these days, such coefficients are still widely used in sociology and make a useful bridge to a topic we will deal with later in this chapter.

Rather than referring to the effect of one variable on another, when dealing with only two variables social scientists often talk about the 'correlation' between them, generally denoted by the letter 'r' (or sometimes 'R'). R stands for ‘relationship’ and talking about relationships between variables does not require us to distinguish between dependent and independent variables. Two variables are related if their values tend to move together (taller people tend to be heavier so there is a relationship between height and weight). There is also said to be a relationship – a negative relationship – if two variables tend to move inversely (the thicker the clouds the dimmer the daylight). If both variables are scaled between 0 and 1 (or, in general, both are measured on the same scale) then measures of correlation will take on approximately the same values as the effects we have been talking about. The effects of 0.4 and 0.67 we just referred to would correspond to correlations of 0.4 and 0.67, or very nearly. Correlations are preferable for some purposes, however, because the value of a correlation coefficient does not depend on the scale of measurement of the variables concerned. If we were investigating the relationship between age in years and income in Euros, the effect of age on income would certainly be far greater than 1.0 and would be hard to interpret, whereas the correlation coefficient would be somewhere in the range –1.0 to

+1.0, just like the coefficients we have been discussing. Such correlations would be interpreted in very much the same way as the effect coefficients we have been describing.

Table 7 shows the approximate substantive meaning to be ascribed to correlation coefficients of different magnitudes when using individual-level and aggregate data. As stated earlier, it is really hard to find strong relationships using individual-level data because those who design the question categories are generally not those who answer the questions, so that any number of misunderstandings can result. Also, individuals are very often quite

Table 7

Strength of correlation	Interpretation with Individual-level data	Interpretation with Aggregate data
r = 0.00 to 0.19	Trivial	Trivial
r = 0.20 to 0.34	Moderate	Slight
r = 0.35 to 0.49	Strong	Moderate
r = 0.50 to 0.65	Spectacular	Strong
r = 0.66 to 0.80	Highly spectacular	Very strong
r = 0.81 to 0.95	Suspect	Spectacular
r = 0.96 to 1.00	Very suspect	Very suspect

unsure about how to answer even questions that they correctly understand. This results in a great deal of error that is largely absent from aggregate data (or averaged out when individual-level information is aggregated). Thus we expect stronger correlations (and stronger effects) with aggregate data than with individual-level data. Indeed, individual-level correlations above 0.8 are so unusual as to generally suggest that something about the analysis was done wrongly, or something about the data is not quite right. Very often in such cases the analyst has employed two variables that are in reality different measures of the same thing. With aggregate data, correlations above 0.9 are quite attainable (though unusual). Only correlations above about 0.95 suggest the testing of tautological relationships.

How widespread?

The extent to which a relationship is widespread is a matter of the number of situations in which it is found. A relationship found only where there are extensive social networks is less widespread than one found also where social networks are absent. Establishing how widespread is a relationship requires the use of multiple independent variables in order to

specify the different circumstances in which that relationship does or does not hold. In the Putnam example we started with, the relationship between networks and governance held only in cases where entrepreneurship was present. So this relationship proved not to be as widespread as originally supposed by Putnam. A relationship that holds only in certain circumstances is said to be subject to an 'interaction'. In this case there was an interaction between entrepreneurship and the extent of social networks such that each had its effect only in the presence of the other. In order to test for interaction effects it is necessary to employ a great many independent variables, one for each of the circumstances in which an effect might or might not be found to hold true. But we need multiple independent variables for an additional reason as well, to which we now turn.

Multivariate analysis

So far, except when examining the Putnam thesis, we have been talking only about so-called 'bivariate' relationships: relationships that may be found when a single dependent variable is examined in relation to a single independent variable. It is unusual to be able to explain much about the world with bivariate relationships, partly because there is so much error in our data (especially in our individual-level data) – error that often needs to be measured and specified in order to correctly estimate the effects of the variables of interest. More importantly, the social world is a complex place. All the circumstances that might affect how widespread a relationship is (as just explained) may also contribute to an explanation of the behaviour of the dependent variable of interest. When we bring in additional independent variables in this way we are said to be ‘elaborating’ our explanation, as mentioned in the opening paragraph of this chapter. Indeed, the need to take account of multiple simultaneous effects on a dependent variable is in practice much more frequent than the need to take account of interaction effects. But as soon as we move beyond bivariate analysis we need new tools for thinking about relationships, and when we use such tools we are said to be performing 'multivariate analysis'.

Strictly speaking, the analysis we performed in Tables 3 and 4 were multivariate analyses because more than a single independent variable was involved. However the tools we used (tables, percentages, percentage differences) were the tools of bivariate analysis. When we move to multivariate analysis proper we need to think of effects in terms of equations, and this is another daunting step that leaves many social scientists with spots before their eyes – unless it is explained to them that equations are perfectly straightforward

tools that everyone uses implicitly every time they add up the charges they expect to incur on their next mobile phone bill.

A typical mobile phone bill has a total that is the result of taking a standing monthly charge and adding to it an amount for calls in excess of some maximum, perhaps an amount for roaming, perhaps an amount for taxes, and so on. The result is a sum that can be spelled out as an equation such as the following:

$$\text{Total due} = \text{standing charge} + \text{charge}_{\text{per excess minute}}*\text{minutes} + \text{charge}_{\text{per roaming minute}}*\text{roaming}$$

(perhaps with another component for taxes). In the equation the plus sign signifies addition and the asterisk signifies multiplication. What one does in order to calculate the total due is multiply the number of excess minutes by the charge per excess minute and the number of roaming minutes by the charge per roaming minute and add those two products to the standing charge. This does not give many people much trouble. What gives them trouble is when the words used in the above equation are replaced with symbols, as in the following:

$$Y = a + b_1X_1 + b_2X_2$$

Here the total due is replaced by the letter 'Y', the standing charge by the letter 'a', the number of excess minutes by the letter 'X' with a subscript of 1 and the number of roaming minutes by the letter 'X' with a subscript of 2. Each 'b' is the charge for the corresponding number of minutes (again with the appropriate subscript).

The use of symbols in place of words looks quite cumbersome but is actually very powerful. By convention we always use the letter Y to stand for the dependent variable and the letter X (with different subscripts) to stand for different independent variables. Each b measures the effect of the relevant X on the dependent variable. The letter a is always used to denote a constant (which might be zero). Evidently we can extend the equation with many more Xs without running out of space on the line, and we can talk conceptually about what we are doing without having to use any specific examples of actual variables. In the Putnam example, we could write the equation that we were implicitly evaluating exactly as above, where Y stands for quality of democratic governance, X_1 for extent of social/political networks and X_2 for availability of entrepreneurship. In practice in this example the a term was zero because quality of governance was zero in the absence of the two necessary

conditions. Note that we cannot actually estimate the effects inherent in Table 2 without considerable gyrations (the only effect we calculated – the 0.67 effect of networks on quality in the case where entrepreneurship was available – is what is called a 'partial' effect, an effect that applies only in a specified circumstance).

In order to calculate effects of independent variables on dependent variables in a multivariate analysis several methods are available, but the most widely used is called 'regression analysis'.

Regression analysis

This type of analysis gets its name, in a most unlikely fashion, from the fact that it was developed by geneticists to study the way in which offspring who are taller or shorter than their parents tend to themselves have children whose height 'regresses towards the mean'. In this brief introduction there is no need for us to explain how the calculations are performed. All that is necessary is to know that, for any given dependent variable, Y, regression analysis produces values for the constant term, a, and for each of the bs used in investigating the relationships concerned. The analyst must supply the data for Y and each of the Xs which will generally be contained in a data matrix such as the one presented earlier, in Table. Using those data, we can estimate the level of turnout at European Parliament elections for different countries quite accurately from the length of time until the next national elections (EP elections gain in importance as 'barometers' public opinion as the next national election approaches), along with a correction for compulsory voting (countries with compulsory voting see much less drop-off in turnout at European Parliament elections than other countries do) and another correction for countries facing their first EP elections (initial, or 'founding' elections tend to see higher turnout). The results can be expressed in terms of an equation that looks like

EP turnout = 56.2 + 29.9*compuls – 3.6*yrsleft + 9.3*firstEP

We will see that both yrsleft and firstEP in practice are interacted with whether the country is a compulsory voting country or not, since compulsory voting countries do not see variations in turnout on the basis of years until the next national election or at founding elections. However, taken at face value this equation tells us that turnout at European Parliament varies around a baseline of 56.2 percent, depending on whether the country has compulsory voting

(which adds nearly 30 percent), how many years there are left until the next national election (turnout is lower by 3.6 percent for each year still to go), and whether this was the first European Parliament election held in the country (see Mark N. Franklin, "How Structural Factors cause Turnout Variations at European Parliament Elections," *European Union Politics* (2:3) 309-328.).

Regression example

In this section we will walk through the analysis that led to the results shown in that equation. The idea that turnout at European Parliament elections would depend solely on features of those elections is not perhaps the first thing that would come to mind when theorizing about EP elections. Much more intuitively appealing would be the thought that turnout at these elections should reflect turnout at national elections, but with some sort of a discount (and with a correction to account for higher turnout in countries with compulsory voting). Table 8 presents the findings of a series of different regression analyses (described as 'models' in the table), each one using slightly different independent variables, in order to step the reader through the findings that led to the rejection of the intuitively more appealing theory (in both of two different variants) and the acceptance of a model (that might be quite surprising to some) that makes no use of national election turnout as an independent variable.

The table is laid out in a fashion customary in contemporary journal articles, with the names of the independent variables down the left-most column and then two columns of coefficients for each model. The first column for each model contains the coefficients of

Table 8

Independent variables	Model A b s.e	Model B b s.e	Model C b s.e.	Model D b s.e.
National turnout	0.5 (0.2)*	0.2 (0.2)	0.1(0.2)	
Compulsory voting	30.0 (3.4)**		25.9 (7.2)**	29.9 (3.0)**
National * Compuls		0.4 (0.1)**	0.1 (0.1)	
Yearsleft * NotCompuls			-3.4 (0.9)**	-3.6 (0.8)**
FirstEP * NotCompuls			8.8 (2.9)**	9.3 (2.8)**
(Constant)	14.6 (14.3)	22.2 (14.0)	47.1 (12.9)**	56.2 (2.5)**
Corrected R^2	0.7	0.7	0.8	0.8
N	64	64	64	64

Note: p= *0.05. **0.01.

primary interest – coefficients that could be transferred to an equation such as the one above (indeed, that equation contains the b coefficients taken from Model D). The second column of

each model is headed s.e. (which stands for 'standard error') these coefficients (between parenthases) measure how much error there is in each b coefficient. It is not important for the purposes of this chapter for you to understand these coefficients, but they are used to determine the level of significance of the effect (generally indicated by one or more stars following the coefficient). If the coefficient is more than about twice its standard error it gets one star to show that it is significant at the 0.05 level. If it is above about three times the size of its standard error it gets two stars to show that it is significant at the 0.01 level. We stand only a 1 in 100 chance of being mistaken when we assert that effects with two stars are real (5 in 100 for effects with one star).

In the last two rows for each model are presented the number of cases included in the analysis (N) and the R-squared associated with the analysis. The R-squared, not surprisingly, is the square of R (or r) – the coefficent discussed earlier that is often used to describe bivariate relationships. The value is squared in multivariate analysis partly because, with more independent variables, it is easier to achieve a high value of R, By squaring this number one arrives at a smaller coefficient more appropriate for use in multivariate analysis (the square of a proportion is a smaller proportion – for example, a half of a half is a quarter so 0.5*0.5 = 0.25). One can use the table (Table 7) for interpreting values of r pretty much as it stands for interpreting values of R-squared when this coefficient is derived from multivariate analysis, distinguishing between individual-level and aggregate data in just the same way as in the bivariate situation.

On the basis of this rather minimal introduction we can proceed to explain why the intuitively more appealing hypothesis (that turnout at EP elections would depend on turnout at national elections) was rejected in favor of an hypothesis that does not even mention national elections. You may find it useful to make a copy of Table 8 to keep by you as you read.

In Model A we do see significant effects of both independent variables, with turnout at EP elections being shown to have a floor of 14.6 percent (though this coefficient is no larger than its standard error which means that it is not significantly different from zero) with an additional component that is half the level of turnout at the most recent national election, and with a correction for compulsory voting that adds a further 30 percent to turnout in compulsory voting countries. Both these coefficients are significant, but the effect of the most recent national election is only significant at the 0.05 level and the variance explained is not stellar for an analysis using aggregate data. Because it is not clear exactly how one would expect compulsory voting to operate in modifying the relationship between national and EP

elections, a second model was then tried (Model B) in which compulsory voting was replaced by an interaction with national turnout. The intuition here was that the effectiveness of compulsory voting laws varies from country to country, so rather than adding the same amount for all compulsory voting countries one should perhaps add a proportion that depends on turnout at national elections in such countries. This formulation, however, works no better than the first forumulation. Again only one of the effects is significant at beyond the 0.05 level and R-squared is not improved.

Model C is an exploratory model that includes every available variable. It is used just to see what happens when other characteristics of the EP election are included in the same model. In this model neither of the variables relating to national elections proves to be significant, though all the other variables are significant at a high level. The R-squared is also much increased, verging on the spectacular (see Table 7). Because the presence of variables whose effects are not significant will nevertheless bias the coefficients of other variables (for reasons we will not go into here), the final model in Table 8 (Model D) is what is called a 'reduced form' model that includes only the variables found significant in Model C. This model explains no more variance than Model C, but it is more parsimonious and its effects are not contaminated by the presence of unnecessary variables. It is thus the model whose values were employed for the equation that was presented earlier.

The way forward

This chapter has provided only the most cursory introduction to quantitative analysis. It can be seen as a shortcut through a forest of methodological approaches with very few detours to take in the variety of possible approaches. Different authors might have chosen different shortcuts, but no-one could cover much more detail in a chapter of this length. This means there is much still to learn about quantitative analysis. In particular, there are a great many other types of multivariate analysis, appropriate to specialized research situations, with the choice between them being largely dictated by the nature of the data being analysed. For example, data in which the cases are different points in time require a whole set of specialized procedures, as do data measured at different levels of aggregation...

Nevertheless, regression analysis is something of an 'industry standard' for multi-variate analysis. Being able to understand the coefficients presented in published research papers that derive from regression analysis (together with the vocabulary used to describe those coefficients and the analyses that give rise to them) will take budding social scientists a

long way. Being able to actually 'do' regression analysis in their own research will help them to be critical consumers of such research findings. Such relatively straightforward skills will also cover a large majority of the situations they are liable to encounter in the world of quantitative research.

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